

Maser Based on Cyclotron Resonance in a Decaying Plasma

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The features of generating electromagnetic radiation in a two-level cyclotron maser whose active medium is a decaying nonequilibrium plasma confined in a magnetic field with the mirror configuration have been examined. It has been shown that, even in the absence of a continuously acting source of nonequilibrium particles (inversion of the medium), the system can exhibit the regimes of the quasimonochromatic or pulse generation of radiation owing to a fast monotonic decrease in the instability threshold that is characteristic of the plasma decay. The theory is confirmed by the results of observations of the burst activity of the decaying pulsed-discharge plasma under the conditions of the electron cyclotron resonance in the direct axisymmetric magnetic trap.

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1. The maser mechanism of generating electromagnetic radiation under the conditions of electron cyclotron resonance in the magnetoactive plasma has long successfully been used to explain a rich class of phenomena in the magnetospheres of the Earth and planets [1–4], in solar coronal loops and in the atmospheres of stars [2, 5], and in laboratory magnetic traps [2, 6–8].

The plasma in most cases is confined in the magnetic field with the mirror configuration. The cyclotron instability is due to the anisotropy of the distribution function of high-energy particles, which is typical for such traps, where the kinetic energy transverse with respect to the magnetic field prevails over the longitudinal component. As a rule, the amplification of electromagnetic waves per passage through the active region is small; for this reason, modes for which multipassage amplification is realized become unstable first. This is possible if radiation is captured in the trap; in this case, the plasma resonator is formed for the working mode. At the nonlinear stage, the increase in the intensity of the electromagnetic field can be limited either by the quasilinear or nonlinear relaxation of the distribution of resonant particles or by the nonlinear properties of the reflection of waves. The emission of electromagnetic waves is accompanied by the pulse ejection of high-energy particles from the trap because they fall within the loss cone. Owing to the sharp decrease in the free energy, the system becomes under the threshold of cyclotron instability; after that a comparatively slow preparation of the next burst (accumulation of high-energy particles) begins and the process is repeated. In this way, the regime of the pulsed generation of broadband pulses is realized. Under other conditions (at higher powers of the source of nonequilibrium parti-

cles), the stationary generation regime of quasi-monochromatic radiation, when the system is always near the instability threshold, and the accumulation phase and discharge of high-energy particles are inseparable. These problems were discussed in detail in, e.g., [3].

In this work, another possibility of generating radiation is realized in the absence of a continuous source of nonequilibrium particles (or, in quantum terms, external pumping sustaining the medium inversion) at the decay stage of the plasma containing two electron components: the relatively cold dense isotropic basic component and the hot anisotropic component with a comparatively low density. This possibility is owing to the difference between the lifetimes of two components. The density of the basic component decreases rapidly and the efficiency of the absorption of electromagnetic waves decreases with it. The confinement time of the anisotropic hot component is much longer and, correspondingly, the growth rates of cyclotron instabilities, which are caused by the presence of the hot component, decrease slowly as compared to the decay of the basic component. For this reason, the instability growth rate in the process of the plasma decay can become larger than losses caused by the absorption of waves in the basic plasma and the generation of electromagnetic radiation becomes possible.

The important role of the basic component of the plasma as an absorbing element that can be responsible for pulsed phenomena in magnetic traps was pointed out in [7]. Similar concepts were discussed previously in application to the dynamics of optical lasers with smooth variation in losses in a cavity [9–11]. However, although the theories of these systems are similar to each other, the set of small parameters of the laser sys-

tems differs from that for the system considered in our work (in particular, the generation threshold is determined by an external inversion source).

2. We consider the simplest, most universal model that describes the quasilinear stage of the development of the instability of a certain proper cyclotron mode in the mirror magnetic configuration. The interaction of resonant electron with the electromagnetic wave exponentially increasing at the linear stage reduces its transverse energy; as a result, some hot electrons fall within the loss cone and leave the trap. These losses reduce the instability growth rate and, finally, restrict the increase in the electromagnetic energy density in the system. The joint evolution of the hot-electron density N and the density of the electromagnetic energy E corresponding to the working mode can be approximately described by the balance equations

$$dN/dt = -\kappa EN, \quad (1)$$

$$dE/dt = (\gamma - \nu)E, \quad \gamma = hN. \quad (2)$$

The first equation describes the rf-field-induced losses of hot particles; other loss channels are disregarded. The second equation describes the increase or decrease in the field in terms of the relation between the increment $\gamma = hN$ and decrement ν on the right-hand side. The coefficients κ and h determine the losses of hot particles and the growth rate averaged along the enhancement line, respectively. The derivation of this system from the kinetic equation of quasilinear plasma theory and comprehensive discussion of its application limits can be found, e.g., in [3].

To calculate the coefficients κ and h , it is necessary to know the solution of the quasilinear kinetic equation for the distribution function of the hot electrons that corresponds to the unstable mode. Nevertheless, it is possible to obtain a simple relation between the parameters κ and h by using the energy conservation law in the system

$$\frac{d}{dt}(\eta W + E) = -\nu E, \quad (3)$$

where W is the energy density of the hot components and η is the efficiency of the maser [7], which reaches a limit value of $2/3$ for sufficiently large anisotropy. The quantity ηW depends on the features of the distribution function; however, it can be estimated as $\eta W \approx NT_h$, where T_h is the effective (taking into account the efficiency) hot-electron temperature. Under the assumption that $T_h = \text{const}$, the relation $\kappa \approx h/T_h$ follows from Eqs. (1)–(3).

When the first equation includes a source of particles, the system of Eqs. (1) and (2) describes the classical regimes of generating radiation in maser systems [3]. As a rule, the parameters h and ν are either constant or periodically time modulated. In our case, the source of active particles is absent, but the parameters h and ν determined by the background plasma can monotonically

vary in time. The last property is responsible for substantial features in the operation of the cyclotron maser.¹ Let an electromagnetic pulse be emitted at a certain time and the system occur in the state below the generation threshold, which is characterized by the low electromagnetic energy and low particle losses. In the process of plasma decay, the decrement of electromagnetic waves decreases and, as a result, the system at a certain time again appears above the generation threshold. The energy density of the electromagnetic field begins to increase and, since this process takes a finite time, the losses of hot particles begin with a certain delay. As a result, a significant amount of electromagnetic energy can be accumulated in the system before the rf-field-induced losses of hot particles transfer the system below the generation threshold, after which the process is repeated. Thus, the quasiperiodic series of electromagnetic-radiation bursts or, if the modulation depth of the radiation intensity is small, a quasimonochromatic pulse can be generated.

3. This qualitative pattern is confirmed by the analysis and examples of numerical solution of the system of Eqs. (1) and (2), which are presented in Figs. 1 and 2. For definiteness, we consider the simplest case, where

$$h = \gamma_0/N_0 = \text{const}, \quad \nu = \nu_0(1 - \nu^*t). \quad (4)$$

Here, N_0 , γ_0 , and ν_0 are the hot-electron density, the growth rate, and the decrement at the initial time, respectively, and ν^* is the relative rate of decreasing the decrement. It is easy to see that, after the renormalization $N \rightarrow N/N_0$, $E \rightarrow E/N_0T_h$, $t \rightarrow \gamma_0t$, the system of Eqs. (1) and (2) depends on three independent parameters: the frequencies ν_0/γ_0 and ν^*/γ_0 and the initial energy density $\epsilon_0 = E_0/N_0T_h$.

Figure 1 shows the solution characteristic of the linear relaxation regime, when the modulation depths of the radiation energy density, $\delta E = E - \bar{E}$, and particle density, $\delta N = N - \bar{N}$, are small. The average particle and energy densities can be found from the condition that the system of Eqs. (1) and (2) is close to the insta-

¹ It is worth noting that the change $N' = N + (\nu_0 - \nu)/h$ reduces the system of Eqs. (1) and (2) to a similar system with the effective particle source $J = -d(\nu/h)/dt + (\nu_0 - \nu)E/T_h$ and constant absorption $\nu_0 = \nu(t_0)$. The second term in J is small at the initial stage [under the condition $(\nu_0 - \nu)E/T_h \ll -d(\nu/h)/dt$] and, when the modulation of E is strong, in pauses between bursts. If this term is disregarded and the relation $J \approx -d(\nu/h)/dt \approx \text{const}$ is assumed, it is possible to use the known results for relaxation oscillations in the two-level maser with the constant particle source [3]. The below analytical analysis and numerical calculations confirm that this approximation makes it possible to correctly determine the period, initial modulation depth, and decrement of relaxation oscillation; however, the long-term evolution of the amplitude of the waves in the pulse is determined incorrectly in this case.

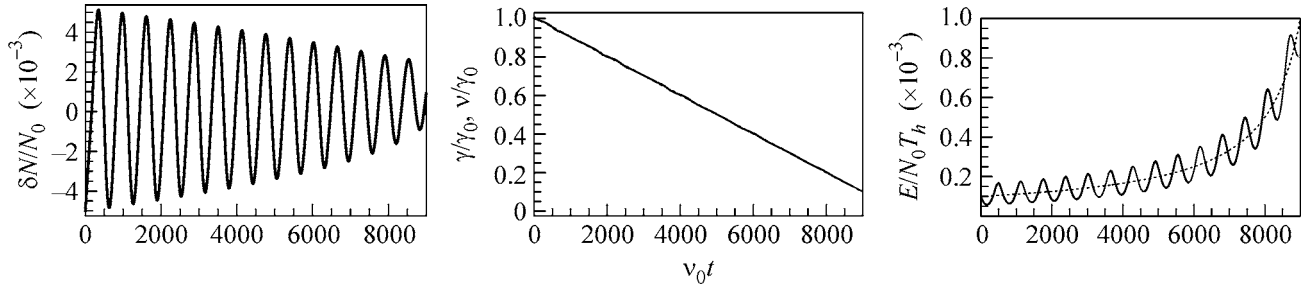


Fig. 1. Time dependence of (left panel) the deviation $\delta N = N - \bar{N}$ of the density of hot electrons from the average value, (middle panel, solid line) increment γ , (middle panel, dotted line) the decrement ν of the working mode of the maser, (right panel, solid line) the density of the electromagnetic energy E , and (right panel, dotted line) the average value $\bar{E}$ of this energy that are obtained for the linear regime by solving the system of Eqs. (1) and (2) with $h = \text{const}$, $\nu = \nu_0(1 - \nu^*t)$, $\nu_0/\gamma_0 = 1.005$, $\nu^*/\gamma_0 = 10^{-4}$, and $\epsilon_0 = 0.99\nu^*/\gamma_0$.

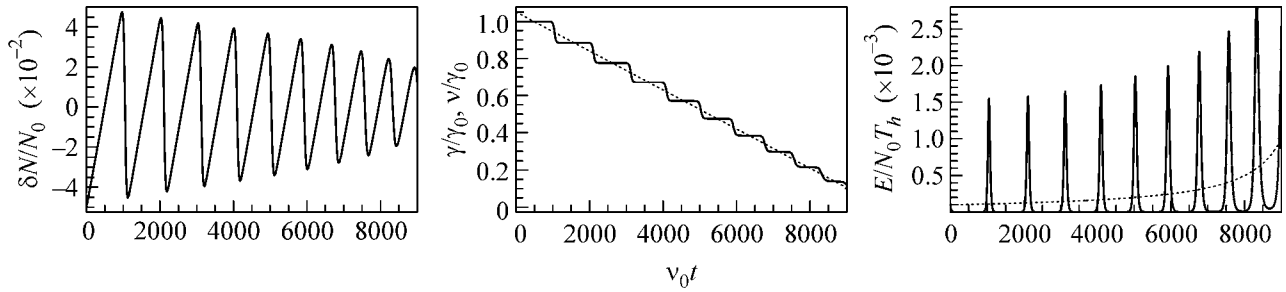


Fig. 2. Same as in Fig. 1, but for the nonlinear regime for $\nu_0/\gamma_0 = 1.05$.

bility threshold condition $\gamma = \nu$ (the γ and ν plots are close to each other in Fig. 1):

$$\begin{aligned} \bar{N}(t) &= \frac{\nu}{h} = \frac{\nu N_0}{\gamma_0}, \\ \bar{E}(t) &= -\frac{1}{\kappa \bar{N}} \frac{d\bar{N}}{dt} = N_0 T_h \frac{-\dot{\nu}}{\gamma_0 \nu}. \end{aligned} \quad (5)$$

Here, $\dot{\nu} = d\nu/dt < 0$. Thus, a variation in the losses leads to the appearance of a new dynamic equilibrium state of the maser with $\bar{E} > 0$. The linearization of the initial equation near this equilibrium state yields

$$\frac{d}{dt} \delta N = -\nu \frac{\delta E}{T_h} + \frac{\dot{\nu}}{\nu} \delta N, \quad \frac{d}{dt} \frac{\delta E}{T_h} = -\frac{\dot{\nu}}{\nu} \delta N. \quad (6)$$

According to the first of these equations,

$$\delta E = -T_h \frac{d}{dt} (\delta N / \nu). \quad (7)$$

The substitution of this expression into the second of Eqs. (6) yields the equation of the harmonic oscillator with a frequency of $\omega = \sqrt{-\dot{\nu}}$:

$$\frac{d^2}{dt^2} (\delta N / \nu) + \omega^2 (\delta N / \nu) = 0. \quad (8)$$

The amplitude and phase of the steady-state oscillations are determined by the initial conditions. Under the assumption of the adiabatically slow variation of the frequency, the oscillation amplitudes are given by the expressions

$$\delta \tilde{N} = N_0 \sqrt{\left(1 - \frac{\nu_0}{\gamma_0}\right)^2 + \left(\frac{\omega_0}{\gamma_0} - \frac{\epsilon_0 \nu_0}{\omega_0}\right)^2} \frac{\nu}{\nu_0} \sqrt{\frac{\omega_0}{\omega}}, \quad (9)$$

$$\delta \tilde{E} = (\omega / \nu) \delta \tilde{N}. \quad (10)$$

Here, ω_0 is the oscillation frequency at the initial time. The linear-approximation condition $\delta \tilde{N} \ll \bar{N}$ is reduced to the condition

$$(\gamma_0 - \nu_0)^2 + (\omega_0 - \epsilon_0 \nu_0 \gamma_0 / \omega_0)^2 \ll \nu_0^2 (\omega / \omega_0). \quad (11)$$

In the case of the linear time dependence, the absorption decrement (4) vanishes in a finite time $1/\nu^*$; in this case, the frequency of small oscillations is a constant equal to $\omega = \sqrt{\nu_0 \nu^*}$. From this condition, the

number of complete oscillation periods in the system in the time $1/v^*$ can be determined as

$$M = \omega/2\pi v^* = \sqrt{v_0/v^*}/2\pi. \quad (12)$$

It is possible to verify that the period of the nonlinear oscillations in the system under consideration is always larger than its linear value; for this reason, Eq. (12) provides the estimate for the maximum possible number of oscillation periods for these parameters. For the possibility of $M > 1$ oscillations existing in the system, the decrement must decrease slowly in time: $v^* < v_0/(2\pi M)^2 \ll v_0$.

When inequality (11) is not satisfied, relaxation occurs in the nonlinear regime for which the generation of pulsed radiation bursts are accompanied by the intense precipitation of hot electrons. Figure 2 shows an example of nonlinear relaxation. Let us analyze the characteristics of a strongly nonlinear case, where the radiation burst duration τ_p is much shorter than the nonlinear period τ_T between sequential bursts. It can be shown that, when the decrease in the decrement with time is adiabatically slow, $-\dot{v} \ll v^2$, the average densities of hot electrons and electromagnetic energy are still determined by linear expressions (5). Indeed, the expression for the electron density follows from an obvious “geometric” analysis, see the average plot in Fig. 2: the density decreases “stepwise,” and the system passes through the threshold $\gamma = v$ (i.e., $N = \bar{N}$) twice in a period at the times when the field intensity is minimal and maximal. The simple relation

$$\Delta N/\bar{N} = \Delta v/v \approx \dot{v}\tau_T/v. \quad (13)$$

between the number of particles precipitating per period, the decrement change per period, and the oscillation period follows from this property. If another point is not indicated, all the time-dependent quantities are taken at the time corresponding to the maximum field intensity (when $\gamma = v$). The energy conservation law given by Eq. (3) per period is written as

$$\Delta NT_h + \Delta E = - \int_{t-\tau_T/2}^{t+\tau_T/2} v E dt. \quad (14)$$

Let us neglect the *minimum* field level $\Delta E \ll \Delta NT_h$ and use the condition of a short pulse $\tau_p \ll \tau_T$, which allows one to take v outside the integral:

$$\Delta NT_h \approx -v \int_{t-\tau_T/2}^{t+\tau_T/2} E dt = -v\tau_T \bar{E}. \quad (15)$$

Comparing Eqs. (13) and (15), we arrive at the expression $\bar{E} \approx -\bar{N} T_h \dot{v}/v^2 = -N_0 T_h \dot{v}/\gamma_0 v$, which exactly corresponds to linear Eq. (5).

If $\tau_p \ll \tau_T$, the change in the decrement v during a short field pulse can be neglected. In this case, the sys-

tem of Eqs. (1) and (2) has an approximate integral of motion, which determines the $E(N)$ dependence:

$$E = E_0 + (N_0 - N)T_h + \bar{N}T_h \ln(N/N_0). \quad (16)$$

From this expression, the relation between the amplitude and pulse repetition period is obtained in the form

$$E_{\max} = E_0 + \bar{N}T_h(x/2 - \ln(1 + x/2)) \approx \bar{N}T_h x^2/8, \quad (17)$$

where

$$x = -\Delta N/\bar{N} \approx -\dot{v}\tau_T/v. \quad (18)$$

In this approximation, exactly half the particles lost per period are precipitated to the time when $E = E_{\max}$.

In order to determine the pulse duration, it is sufficient to take into account the time-linear change $\gamma - v \approx \Delta v t/\tau_p$ during the “active phase” of the pulse generation $t \in [-\tau_p/2, \tau_p/2]$. Substituting this expression into Eq. (2), we obtain the pulse shape $E = E_{\max} \exp(-vx^2/2\tau_p)$. The substitution of this pulse into the energy conservation law given by Eq. (15) yields the following equation determining the pulse duration τ_p :

$$x = \frac{E_{\max}}{\bar{N}T_h} \sqrt{\frac{2\pi v \tau_p}{x}} \operatorname{erf}\left(\sqrt{\frac{v \tau_p x}{8}}\right). \quad (19)$$

It is worth noting that, if the fraction of particles is small, $x \ll 1$, it follows from Eqs. (17) and (19) that $v\tau_p x = \alpha$, where $\alpha \approx 12.1$ is the root of the transcendent equation $\sqrt{2\pi\alpha} \operatorname{erf}(\sqrt{\alpha/8}) = 8$. In other words, $\tau_p \tau_T \approx 12.1/(-\dot{v})$. Hence, under our definition, the enhancement of the field energy density during the active phase depends slightly on the parameters of the problem and is relatively small:

$$E_{\max}/E_0 \approx \exp(v\tau_p x/8) \approx e^{\alpha/8} \approx 4.5. \quad (20)$$

For comparison, we also consider the “passive phase” between radiation pulses at which the field minimum is reached. Between the pulses, the density of the hot particles and the instability growth rate can be treated as constants, whereas the decrement can be considered as a linearly varying quantity. As a result, the total growth rate also varies linearly in time, $\gamma - v \approx -\Delta v t/\tau_T$, and this change occurs in the time interval $t \in [-\tau_T/2, \tau_T/2]$ longer than the active phase. Here, we assume that the passive phase continues over the entire generation period $\tau_T \gg \tau_p$. The pulse shape is the Gaussian $E = E_{\min} \exp(vx^2/2\tau_T)$, and $E = E_0$ at $t \approx \pm\tau_T/2$ (at $E = E_0$, the solutions are matched at the interface between the active and passive phases). By analogy with Eq. (20), one can determine the field enhancement

$$E_0/E_{\min} \approx \exp(v\tau_T x/8) \approx \exp(3\tau_T/2\tau_p) \gg 1. \quad (21)$$

Crude estimates for $E_{\min}$ and $E_{\max}$ can be obtained by setting $E_0 = \bar{E}$ in Eqs. (20) and (21), i.e., by estimating the “transition” energy density level in terms of the

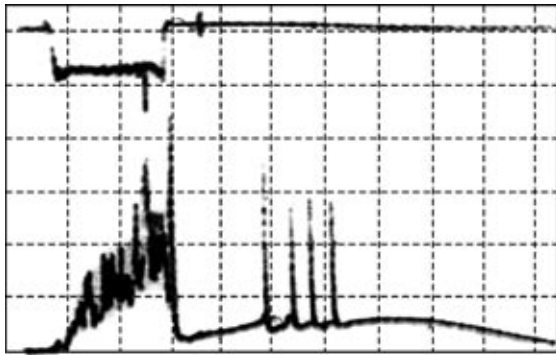


Fig. 3. Oscillogram of (upper ray) the signals of the envelope of a microwave pulse of the gyrotron and (lower ray) the signal from the pin diode of the electron cyclotron discharge in nitrogen. The sweep is equal to 500 μ s per graduation step.

period-averaged value. For more accurate estimates, it is necessary to develop the higher order approximation involving the square time dependence of $\gamma - \nu$ as in [11].

4. In conclusion, we present the first results of the laboratory observation of quasiperiodic bursts of pulsed precipitations of high-energy electrons at the decay stage of the plasma of an electron cyclotron resonance discharge in a direct axisymmetric magnetic trap (the magnetic field in the mirrors is $B_{\max} \approx 1.8$ T, the distance between the mirrors is 25 cm, and the mirror ratio $B_{\max}/B_{\min} \approx 5$). The experiments were carried out with the use of the discharge sustained in nitrogen by the gyrotron with a frequency of 37.5 GHz, a power of 80 kW, and a pulse duration of 1 ms. To measure the flux of high-energy (7–100 keV) electrons from the trap, the pin diode placed on the system axis beyond the trap is used. The key feature of the experiment is the use of a pulsed gas puff, which makes it possible to change (control) the plasma decay after the end of the microwave pulse. The setup and diagnostic methods were described in detail in [8]. We note that this setup was previously used to analyze the cyclotron instability of whistler waves that is developed in a relatively dense plasma during the microwave pulse in the presence of the continuous source of nonequilibrium electrons, which is caused by modification of their distribution function owing to the electron cyclotron resonance interaction with the microwave field.

Figure 3 shows the oscillogram of (upper ray) the signals of the envelope of a microwave pulse of the gyrotron and (lower ray) the signal from the pin diode. It is seen that, in addition to the intense precipitation of electrons during the microwave pulse (acting as a source of nonequilibrium particles that is responsible for the development of the instability of whistler waves), intense short-pulse (with a duration of ~ 5 μ s) quasiperiodic (150–200 μ s) emissions of high-energy electrons are also observed. Such burst emissions of

electrons appear at a certain puff rate of molecular nitrogen with a certain (~ 1 ms) delay after the end of the microwave pulse.

Precipitations after the end of the microwave pulse that are observed after the end of the microwave pulse can likely be explained by the above cyclotron mechanism. Indeed, according to numerous experiments [12, 13] and numerical simulation [14], in addition to the basic cold plasma, the hot anisotropic electron component is formed in the electron cyclotron resonance plasma under the action of the resonant heating. In this experiment, the plasma is formed with a basic-component density of 10^{13} cm^{-3} , an electron temperature of about 300 eV, a hot-electron density of 10^{10} – 10^{11} cm^{-3} , and an effective hot-electron temperature (primarily characterizing the transverse energy) of 10 keV [12]. The emissions of the high-energy electrons that are observed at a noticeable time after the end of the microwave pulse can be attributed to the development of the kinetic plasma instabilities of the “gyrotron” type, which are associated with the excitation of the electromagnetic waves with the quasitransverse propagation with respect to the external magnetic field in the low-density plasma [2]. A substantial feature of such a scenario is that the instability threshold is determined by not only the absorption of waves in the background plasma but also by the transparency of the plasma layer. During the microwave pulse and at the initial plasma-decay stage, such instabilities are suppressed due to the depression of cyclotron radiation in the evanescence region of the overdense plasma, and the development of instabilities becomes possible in sufficiently low-density ($\ll 10^{12}$ cm^{-3}) plasma only after a finite (~ 1 ms) time after the end of the microwave pulse, when (according to estimates) the condition $\omega_{pe} \ll \omega_{ce}$ is satisfied, where ω_{pe} and ω_{ce} are the plasma and cyclotron frequencies of the electrons, respectively. A more detailed description of this experiment and the results of its analysis, which includes the calculation of the dynamics of the plasma parameters in the decay process, the identification and estimation of the growth rates of the potentially instable modes, and the numerical solution of the system of equations of the cyclotron maser will be presented elsewhere.

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